

Lagrange equation review:

Minimal coordinates q_i , $i=1, N$

one for each configuration DoF

$$\mathcal{L} = E_k - E_p = \mathcal{L}(q_i, \dot{q}_i)$$

Lagrange equations (n of them)

$$\textcircled{\text{I}} \quad \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = Q_i$$

$$\textcircled{\text{II}} \quad \text{2nd order constant equations}$$

— Non holonomic cases

Note:

- constant forces for
- holonomic constant don't show up
- conservative forces are in E_p

Q_i are for

- non-holonomic forces
- non conservative forces

What are the Q_i 's?

Method 1

$$dW = (Q_1) dq_1 + (Q_2) dq_2 + (Q_3) dq_3 \dots (Q_i) dq_i$$

↑
in terms of q_i

Method 2:

forces are applied at positions

$$x_j = \begin{cases} x_1 = x_1 \\ x_2 = y_1 \\ x_3 = z_1 \\ x_2 = x_2 \\ \vdots \end{cases}$$

$$Q_i = \int \frac{\partial x_j}{\partial q_i} F_j$$

└ points of force application

or

$$Q_i = \sum \frac{\partial \vec{x}_j}{\partial q_i} \cdot \vec{F}_j$$

Method 3 : only for extra constraint forces

constraint equation in velocity form

$$(A_1) \dot{q}_1 + (A_2) \dot{q}_2 + \dots = f(q_i, t)$$

Work of constraint forces

$$(\quad) \delta q_1 + (\quad) \delta q_2 + \dots = 0$$

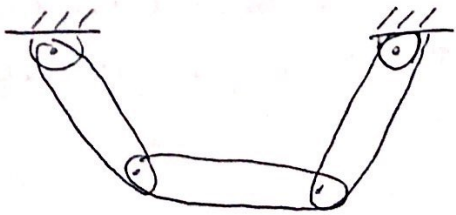
↑
workless constraint

Lagrange multipliers

$$\begin{bmatrix} Q_1 \\ Q_2 \\ \vdots \end{bmatrix} = F \begin{bmatrix} A_1 \\ A_2 \\ \vdots \end{bmatrix}$$

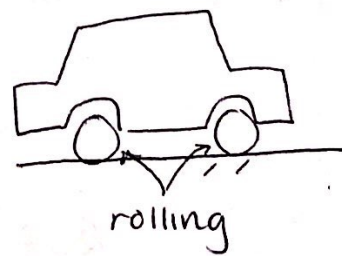
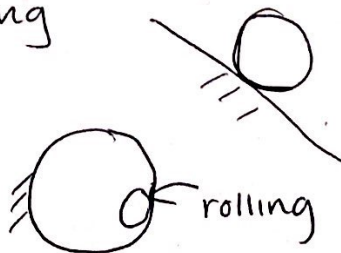
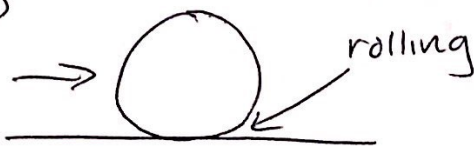
Note: OK to treat holonomic constraints as added constraint

ex)



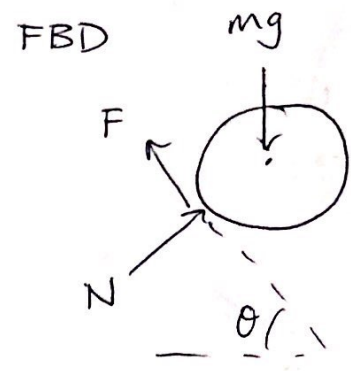
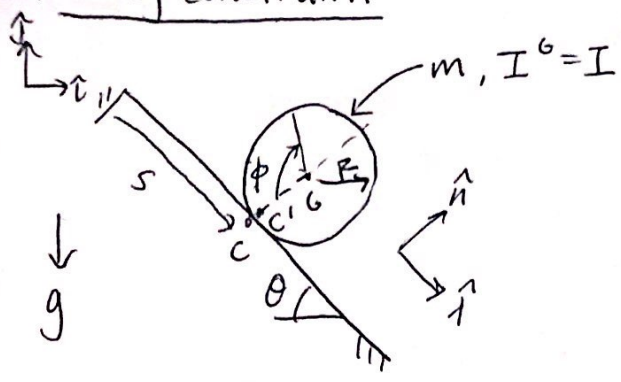
Rolling

ex)



not neglect
inertia
of wheels

Rolling constraint:



Method 1:

$R\phi = s$ ← difficult to keep track of

Method 2:

$\vec{v}_c = \vec{v}_c = \vec{0}$ (Laurie-Andersen equation)
 singer lol. "Let x equal x "

$\vec{v}_G + \vec{\omega} \times \vec{r}_{c/G} = \vec{0}$

$\dot{s}\hat{x} + (-\dot{\phi}\hat{k}) \times (-\hat{n}R)$

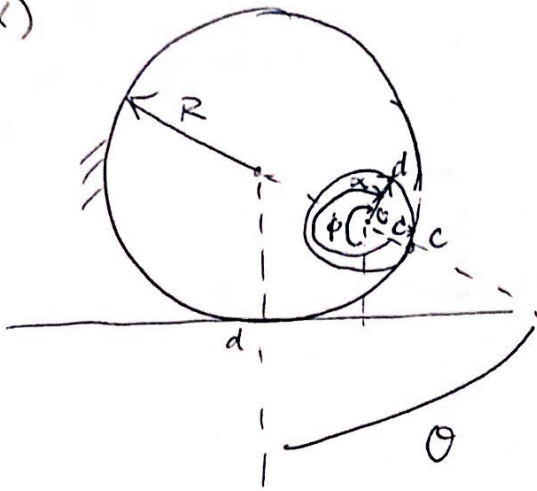
$\frac{d}{dt} \{ \} \Rightarrow \boxed{\dot{s} = \dot{\phi} R}$

$\Rightarrow \boxed{\ddot{s} = \ddot{\phi} R}$

ex)

Relationship between angles

$$\alpha = \phi + \theta$$



Method 1:

• Match arc lengths between little circle + big circle

$$\begin{aligned} \theta R &= \alpha r \\ &= (\phi + \theta) r \end{aligned}$$

Method 2

$$\vec{v}_c = \vec{v}_c$$

$$\vec{v}_B + \vec{v}_{C/B} = \vec{0}$$

$$(R-r)\dot{\theta}\hat{e}_\theta + (-\phi\hat{k}) \times r\hat{e}_r = \vec{0}$$

something